

# Shivayogi Akki

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## PROFESSIONAL SUMMARY

- Ph.D. candidate in Electrical and Computer Engineering at Binghamton University specializing in reinforcement learning, control, and legged/humanoid robotics
- Prior experience in projects like Unitree G1 fault-tolerant locomotion improving two-joint-fault success from 0% to 72%, and PPO-based Go1 control with 41% lower mechanical CoT and 0.25–0.33 s faster disturbance recovery

## EDUCATION

**Binghamton University, SUNY, Thomas J. Watson College of Engineering and Applied Science** **Binghamton, NY**  
Michigan Technological University Houghton, MI  
*Doctor of Philosophy in Electrical and Computer Engineering* *Expected May 2028*  
Dissertation: Legged locomotion for Human and Humanoid robots in varied gravitational environments.

**Michigan Technological University** **Houghton, MI**  
*Master of Science in Mechatronics, Robotics, and Automation Engineering* *Jan 2022 - Dec 2023*

**Visveswaraya Technological University, JSSATE** **Karnataka, India**  
*Bachelor of Science in Mechanical Engineering* *Jun 2017 - Aug 2021*

## TECHNICAL SKILLS

**Programming & Machine Learning:** Stable-Baselines3, Gymnasium, Git/GitHub, Linux, Python, C++, MATLAB, PyTorch  
**Robotics & Simulation:** MuJoCo, ROS, Isaac Gym, RoboGuide, FANUC Robot Programming  
**Hardware & Industrial Automation:** Unitree Go1/G1, Intel RealSense, NVIDIA Jetson Nano, FANUC Robots, Allen-Bradley PLCs

## RESEARCH EXPERIENCE

**Binghamton University, RoLAC Lab** **Binghamton, NY**  
*Research Assistant (Fault tolerant goal oriented robot locomotion)* *Aug 2026 - Sep 2026*

- Developed SAC reinforcement-learning controllers for a 23-DoF Unitree G1 humanoid walking using Python, MuJoCo, Gymnasium, and Stable-Baselines3, increasing simulated success rate under two joint-lock faults from 0% to 72%
- Implemented torque-limiting controls, reducing peak torque by 26% (70 to 52 Nm) versus the uncapped policy with a hip joint locked
- Trained fault-tolerant crawling policies, increasing survival rate with forward progress from 65% to 92.5% under three simultaneous knee/ankle joint locks with hips free in simulation

**Binghamton University, RoLAC Lab** **Binghamton, NY**  
*Research Assistant (Template model based analysis of human locomotion gaits)* *June 2025 - Present*  
*Funded by NASA*

- Built a 9-DOF compliant planar biped and an end-to-end SAC pipeline in MuJoCo/Gymnasium in which walking, aerial running, and skipping all emerged from a single gait-agnostic reward - no gait labels, reference trajectories, or prescribed contact schedule
- Validated frozen controllers on achieving 100% fall-free trials across the three gaits with 0.07–0.26 m/s steady-state velocity-tracking RMSE, 0.44–0.84 mechanical cost of transport
- Developing an 18-muscle actuators musculoskeletal biped to be trained for efficiency optimization in Earth and lunar gravity locomotion

**Binghamton University, RoLAC Lab** **Binghamton, NY**  
*Research Assistant (Learning-Based Quadruped Locomotion and Autonomous Navigation)* *Jan 2025 - Mar 2025*

- Developed a VLM–RL autonomous navigation stack integrating LiDAR, RealSense/YOLO perception, A\* planning, and learned locomotion, achieving 92% task completion across 10 randomized physical Unitree Go1 trials with a 101 s mean completion time
- Evaluated cyber-physical robustness through prompt-injection attacks, demonstrating 90% attack success and a 23 s mean redirection time without modifying VLM weights or training data

**Michigan Technological University, Department of Electrical Engineering** **Houghton, MI**  
*Research Assistant (Benchmarking controls for legged robot locomotion)* *Aug 2023 - Mar 2024*  
*Funded by NSF*

- Developed a standardized MuJoCo benchmarking framework for a 12-DoF Unitree Go1, implementing PPO reinforcement learning and predictive-sampling MPC for 0.5 m/s locomotion; evaluated both controllers under identical disturbance and terrain conditions
- Demonstrated that PPO achieved 41% lower mechanical Cost of Transport (1.768 vs. 2.993) and 0.25–0.33 s faster disturbance recovery, while quantifying controller robustness under perturbations of up to 950 N

**Michigan Technological University, Department of Electrical Engineering** **Houghton, MI**  
*Course Project (Simulation based industrial assembly line using HIL with PLC module)* *Mar 2023 - Apr 2023*

- Integrated an Allen-Bradley PLC with FANUC RoboGuide over EtherNet/IP, configuring PLC/robot I/O for a simulated car-door assembly cell with 3 FANUC R-2000iB/165F robots, 2 conveyor lines, and pick-and-place/welding operations
- Developed RSLogix 5000 ladder logic and a FactoryTalk HMI for remote production-start control and bidirectional PLC–robot communication, enabling hardware-in-the-loop validation of the automated cell prior to physical testing

## PUBLICATIONS

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### Orcid ID: 0009-0001-3502-3590

- **S. Akki** and T. Chen, “Benchmarking model predictive control and reinforcement learning-based control for legged robot locomotion in mujoco simulation,” *IEEE Access*, vol. 13, pp. 108 732–108 742, 2025.  
DOI: 10.1109/ACCESS.2025.3582523.
- Obidov, D., **Akki, S.**, Chen, T., & Yang, K. (2025). Silent sabotage: Internal state triggered backdoor attacks on llm-powered robotic systems. In *Security and Privacy in Cyber-Physical Systems and Smart Vehicles* (pp. 107-121). Cham: Springer Nature Switzerland.  
DOI: 10.1007/978-3-032-33701-6\_7
- Biligiri, D., Smitterberg, J. A., **Akki, S.**, Goodwine, B., & Chen, T. (2025). FractionalNet: a symmetric neural network to compute fractional-order derivatives: D. Biligiri et al. *Nonlinear Dynamics*, 113(24), 33381-33398.  
DOI: 10.1007/s11071-025-11736-6
- N. Rawashdeh, R. Bondalapati, P. Patil, G. S. Ajmani, and **S. Akki**, “Hardware-in-the-loop simulation of a programmable logic controller, industrial robots and conveyor systems using roboguide,” in 2024 22nd International Conference on Research and Education in Mechatronics (REM), 2024, pp. 132–137.  
DOI: 10.1109/REM63063.2024.10735684.